Exploring Image Classification Robustness of ONE-PEACE

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***Abstract*—In this study, we assess the efficacy of ONE-PEACE, an integrative multi-modal representation model encompassing language, vision, and audio, specifically focusing on uni-modal image classification task across three distinct adversarial and out-of-distribution variants of ImageNet. Our evaluation involves utilizing the fine-tuned vision module of ONE-PEACE to evaluate the robustness of the proposed approach. Additionally, we replicate the initial findings on ImageNet-1K and conduct a comparative analysis with state-of-the-art models in the realm of robust image classification. The reported metrics from the original study have been successfully reproduced, contributing to the comprehensive evaluation of ONE-PEACE in this context.**

***Keywords*—computer vision, object classification, image classification, object recognition, robustness.**

I. Introduction

Although the problem definition may look very simple, as it is just assigning categories to a set of images, many hardships are encountered while building successful image classifiers that are robust to various distribution shifts in unseen test data [1] [2] [3]. Previously, numerous methods which were successful in ImageNet image classification task were shown to have poor generalization on test sets that consisted of examples with aforementioned distribution shifts [4] [5].

In this study, we aimed to analyze the robustness of an extensible, multi-modal model ONE-PEACE [6] rather than focusing on uni-modal vision models to show how multi-modal pre-training and shared representations could effect the model robustness to different types of noise, perturbations, distributions, and adversaries. This paper's main contributions involve thoroughly studying the ONE-PEACE method, providing a comprehensive evaluation. Additionally, it compares the method to enhance our understanding and context of ONE-PEACE in the domain of robustness.

The rest of this paper is organized as follows: Section 2 presents preliminaries on image classification and model robustness. Datasets and Methodology parts are proposed in Sections 3 and 4 respectively. Experimental Setup and Results are described in Section 5. Conclusion and future research directions are given in Section 6. The datasets and code used to make these evaluations are all publicly accessible[1].

[1]The following notebook contains dataset download scripts and repository clones to replicate our results: "https://colab.research.google.com/drive/1KtTcItXJxtzYh_7wNGcyKu1DHjoVVMNC?usp=sharing"

II. Related works

A. Image Classification

Many researchers in the realms of image processing and machine learning have extensively explored the domain of image classification from its early stages [7] [8]. There are numerous databases and models for the purpose of image classification in different domains such as medical [9] [10] and e-commerce [11]. A more wide-domain database, ImageNet, is a core part of ImageNet Large Scale Visual Recognition Challenge (ILSVRC), that contains images and objects from various domains, currently providing 21000 [12] ontological categories and is still widely used. This annual challenge and the dataset allows researchers to see breakthroughs in computer vision more clearly over time, a prominent example to this was AlexNet [13], achieving top-1 and top-5 error rates of 37.5% and 17.0% and greatly outperforming the previous state-of-the-art. More recently, based on the advancements in deep neural network architectures and pre-training methods in natural language processing [14], and adaptation of these methods to computer vision [15] [16] led to the current SoTA in ImageNet classification task [17] reporting 95.12% top-1 accuracy.

B. Robustness

The study of how well a system can withstand distribution shifts in the test data has been explored by a significant number of researchers. Robustness can be defined in terms of robustness to adversarial and out-of-distribution examples [4], noise and perturbations [5]. To address this challenge, the research community has recently leaned towards enhancing the "diversity" of training data through the implementation of data augmentation techniques [18] [19]. The underlying hope is that exposing a network to a wide range of diverse image augmentations would increase its likelihood of developing invariance to common corruptions. However, covering the entire spectrum of common corruptions proves to be a challenging task. Other augmentation techniques include mixing up patches in the training images [20] [18], robustness to occlusion by randomly erasing parts of the image [21] which have all been utilized by ONE-PEACE.

III. Datasets

In this section, we provide brief descriptions of the datasets used for the robustness experiments. ImageNet-O [4] is exclu-

ded from the study since it contains images from ImageNet-22K, which is a part of the pre-training data for ONE-PEACE vision branch. We were not able to include other robustness tasks such as ObjectNet [22], ImageNet-C and ImageNet-P [5] due to the time limits. However, this problem is considered for future work. Examples from each dataset is shown in Figure 1 and common statistics are presented in Table I.

Dataset	ImageNet-1K	A	Renditions	Sketch
# of examples	50000	7500	30000	50000
# of categories	100	200	200	1000

TABLO I: Number of examples and categories in each dataset used in this work. Distribution of examples per category are equal for all datasets.

A. ImageNet-1K [23]

A manually labelled object detection and classification dataset containing 1000 object classes and corresponding bounding box annotations. The data collection and annotation methodology is as follows: determine object categories from ImageNet [24], gather a collection of candidate images through multiple search engines, and ultimately refining the millions of images using the crowdsourcing approach developed by [24]. For image classification task, the dataset contains 1.281.167, 50.000, and 150.000 samples for train, validation, and test splits respectively.

B. ImageNet-A [4]

A challenging, adversarial dataset for image classification models that generally perform well on ImageNet dataset. ImageNet-A consists of 200 broad categories from ImageNet-1K. The data is collected and filtered by obtaining weakly labeled images, applying adversarial selection by eliminating images that could not deceive ResNet-50 models, and finally choosing the low-confidence images.

C. ImageNet-Renditions (R) [25]

ImageNet-R features 30,000 image variations across 200 ImageNet classes, same as [4]. The selection considers abundant renditions in classes like "goldfish" instead of harder to find class renditions (eg. "crane") and considers coarse-grained categories since it may not be possible to differentiate fine-grained renditions. Images are sourced mainly from Flickr, using diverse queries like "art," "cartoon," etc. After the collection, the data is annotated by crowd-sourcing and validated by graduate students.

D. ImageNet-Sketch [26]

As the name suggests, this dataset comprises of 50,000 images, with 50 images allocated for each of the 1000 ImageNet classes, but in black and white sketch form. The dataset is created using Google Image queries for "sketch of" followed by the standard class name, focusing on the "black and white" color scheme.

IV. Methodology

A. ONE-PEACE

1) Architecture: ONE-PEACE is a large model with 4B trainable parameters in its 3 modality state, vision branch only contains 1.52B parameters. In order to be able to process signals from different modalities, the model architecture starts with "adapters" for audio, language, and vision. On top of these adapters, "model fusion encoder" structure based on the Transformer architecture [14] is used for the purpose of creating shared representations for these modalities.

2) Pre-training: ONE-PEACE pre-training is performed using two tasks: cross-modal contrastive learning and intra-modal denoising contrastive learning. Authors state that cross-modal contrastive learning imparts the model with the ability for cross-modal retrieval, while intra-modal denoising contrastive learning enhances the model's fine-tuning performance in subsequent tasks. The pre-training is first done one image-text pairs, then audio-text pairs, using text as an anchor between modalities. They use LAION-2B [27] for image-text pre-training and numerous public datasets for audio-text pre-training.

V. Experiments

A. Setup

As we utilize pre-trained and fine-tuned models directly from the original work without conducting any additional training, there are no training hyperparameters involved. However, we use two models with different input sizes ONE-PEACE-384[2] and ONE-PEACE-512 [3]. The models we use are previously pre-trained on LAION-2B [27], further trained on ImageNet-21K, and finally fine-tuned on ImageNet-1K. Both models have dropout rate of 0.0, drop path rate of 0.1, global pooling set to True. To accommodate both the model and the data within the Colab Pro+ VRAM, we employ a batch size of 32 and use an A100 GPU. We run inference with both models on 4 datasets discussed in the paper, and report top-1 accuracy scores for each dataset.

B. Results

All of our experiments and performances of other models on the described datasets are presented in Tables II and III. Experimental results show that even though there is a performance drop in adversarial test sets, ONE-PEACE stays comparable with current state-of-the-art methods due its utilization of various augmentation techniques in the pre-training and fine-tuning phases.

VI. Future work and limitations

This study has limitations concerning the diversity of models and datasets employed, hindering the establishment of a comprehensive robustness benchmark. In the future, we aim to include datasets such as ObjectNet, ImageNet-C, ImageNet-V2 [33], ImageNet-P, include a replication of state-of-the-art

[2]https://one-peace-shanghai.oss-accelerate.aliyuncs.com/one_peace_checkpoints/onepeace_ft_21kto1k_384.pth

[3]https://one-peace-shanghai.oss-accelerate.aliyuncs.com/one_peace_checkpoints/onepeace_ft_21kto1k_512.pth

(a) ImageNet-1K (b) ImageNet-A (c) ImageNet-R (d) ImageNet-Sketch

Şekil 1: Examples from stringray (n01498041) category of ImageNet variants.

Model	Input Size	1K	A	Renditions	Sketch
ONE-PEACE	384	89.6	-	-	-
ONE-PEACE	512	89.8	-	-	-
ONE-PEACE (ours)	384	89.6	83.9	91.6	70.2
ONE-PEACE (ours)	512	89.7	85.0	91.2	70.43

TABLO II: Comparison of replicated top-1 accuracy results on ImageNet dataset.

Model	1K	A	Renditions	Sketch
ResNet-50 [28] [25] [29]	79.3	11.4	37.2	-
ResNeXt-101 [30] [25] [29]	81.2	45.4	75.8	-
Model soups (ViT-G/14) [31]	90.9	92.7	95.5	74.2
Model soups (Basic-L) [31]	91.0	94.2	96.1	77.2
LSMD [32]	78.5	87.8	91.13	89.5
ONE-PEACE-384	89.6	83.9	91.6	70.2
ONE-PEACE-512	89.7	85.0	91.2	70.43

TABLO III: Top-1 accuracy comparison on ImageNet variations. Results other than ONE-PEACE are collected from various studies. Best accuracies are underlined.

methods for robust image classification and further metrics to be able to analyze the results more rigorously. ImageNet-O could also be included for tests given that we do not use ImageNet-21K as our training set by not using previously trained models. Model performances can be analyzed in terms of subcategories in ImageNet-Renditions, where they provide sketches, sculptures, tattoos and more.

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